

Electric 9.07: EDA And Circuit Layout Made Easy

If you are looking for an open source electronic design automation (EDA) software, Electric is one of the highly efficient options. This very-large-scale-integration (VLSI) tool lets you draw schematics and create integrated circuit (IC) layouts digitally. Electric is also compatible with hardware description languages like VHDL and Verilog.

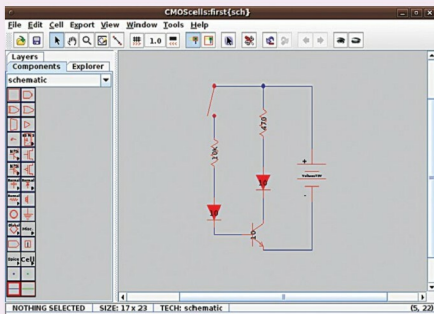
The different design technologies

The default library of Electric software consists of most of the components necessary for IC design, supporting different technologies and multiple file formats. The different categories of circuits that Electric can design include complementary metal-oxide semiconductor (CMOS), n-channel MOSFETs, bipolar technologies based on integrated injector logics, newer layout technologies like thin-film, carbon nanotube and photonics, abstract circuits like digital filters, temporal logics and so on. Each design technology comes out on the interface as a collection of components connected with wires.

A major advantage of the software is that the technology describes the technical parameters of the components and wires, like design rules, simulation aspects and graphical descriptions to name a few. Users can also add modules to custom-design the environment with the help of a technology editor.

Analysis and synthesis properties

Electric has a strong inventory of analysis and synthesis tools that



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create simulation by storing the connectivity and geometry. These include:

Design-rule checking. This tool checks and validates the layout's design rules compliance and displays spacing errors, notch errors, minimum size errors and other violations. Interfaces like Caliber and Assura can be used to check design rules of external circuits.

Electrical rule checking. The electric rule checker has two functionalities: ensure proper electrical contact and spacing throughout the circuit, and check for antenna rule violations.

Simulation. Electric has an integrated 12-state switch-level simulator called ALS and a built-in IRSIM simulator version that displays waveforms in a sepa-

rate window and allows users to cross-probe. Electric also supports a number of simulators which have to be attained exclusively, for example, device-level simulators like Spice, switch-level simulators like Silos, Tegas, ESIM, RSIM, RNL, Cosmos and Mossim, and behavioural simulators like Verilog and Fast-Henry.

Generation. Electric has four in-built generator technologies:

1. PLA generator, which functions from a library of PLA elements, facilitating customised arrays
2. ROM generator, which builds a ROM layout from a personality table
3. Pad frame generator, which places pad cells around a chip core and wires them together
4. Fil generator, which ensures